

AWS Intern – Hands-On Task

Highly-Available Web Application

Submitted By

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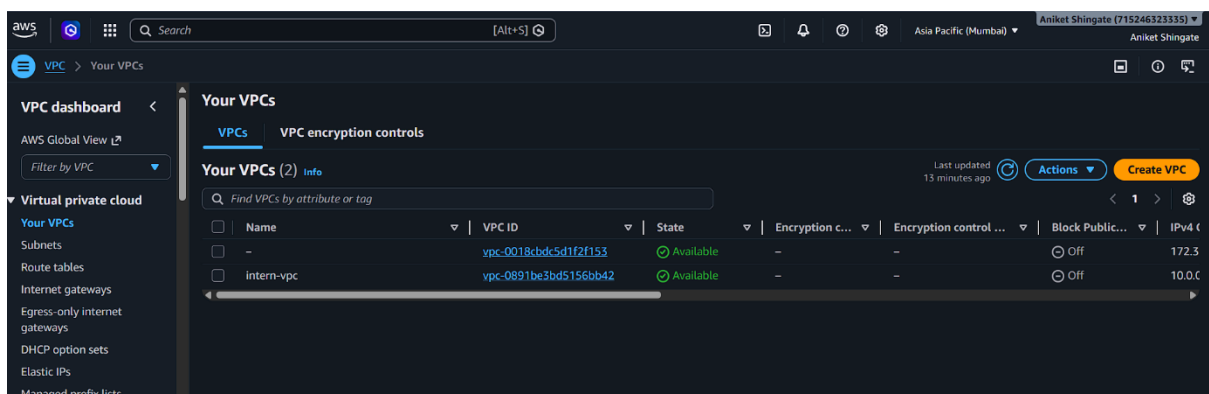
Project Overview

This project demonstrates the deployment of a highly available web application architecture on AWS using the AWS Management Console.

The infrastructure was designed across two Availability Zones using public and private subnets. An Application Load Balancer was configured to distribute traffic across EC2 instances deployed in private subnets through an Auto Scaling Group.

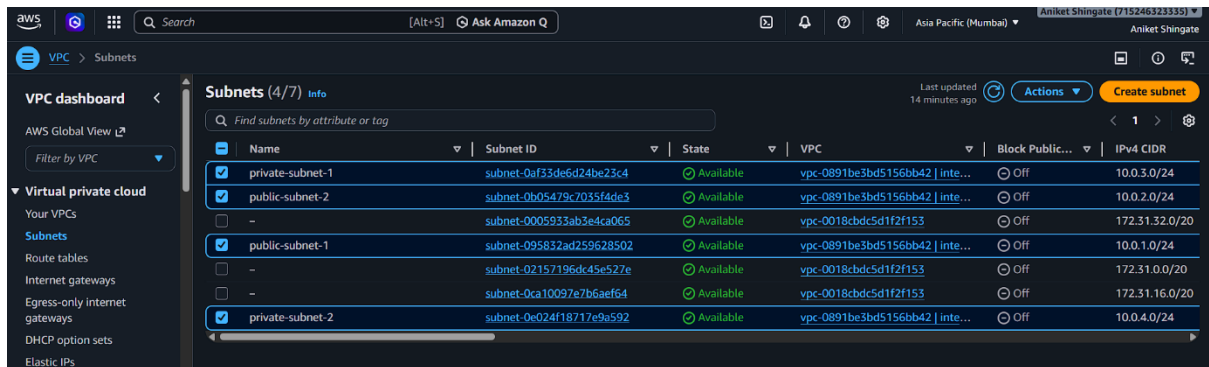
NGINX was installed on the EC2 instances using User Data scripts, and health checks were configured through the Target Group to ensure high availability and self-healing capability.

VPC Configuration:



Created a custom VPC with public and private subnets across two Availability Zones.

Subnet Configuration:

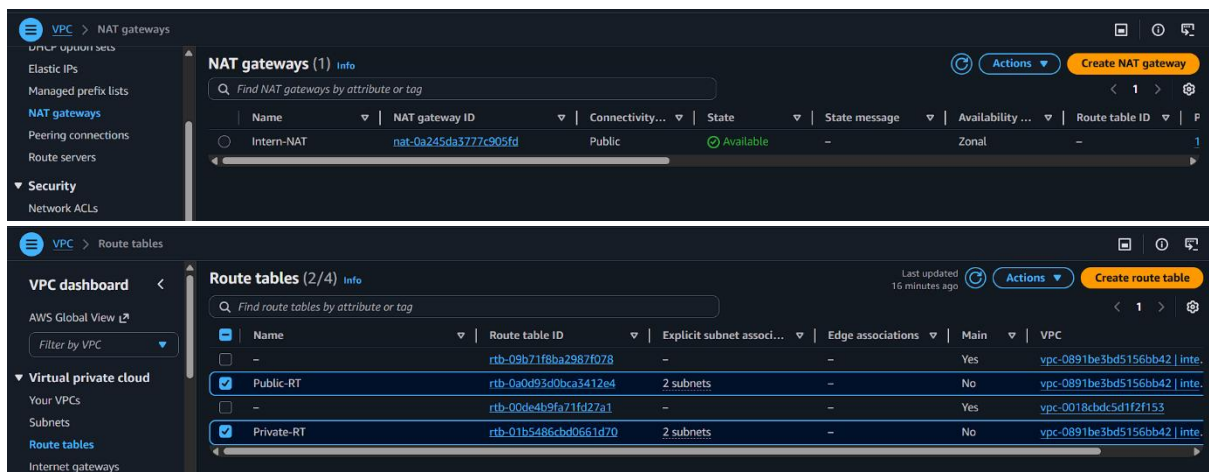


The screenshot shows the AWS VPC console's 'Subnets' page. The left sidebar contains the 'VPC dashboard' and 'Virtual private cloud' sections. The main area displays a table of subnets with columns for Name, Subnet ID, State, VPC, Block Public IP, and IPv4 CIDR. There are four subnets listed: two private and two public, all in an 'Available' state.

Name	Subnet ID	State	VPC	Block Public IP	IPv4 CIDR
private-subnet-1	subnet-0af33de6d24be23c4	Available	vpc-0891be3bd5156bb42 inte...	Off	10.0.3.0/24
public-subnet-2	subnet-0b05479c7035f4de3	Available	vpc-0891be3bd5156bb42 inte...	Off	10.0.2.0/24
-	subnet-0005933ab3e4ca065	Available	vpc-0018cbd5d1f2f153	Off	172.31.32.0/20
public-subnet-1	subnet-095832ad259628502	Available	vpc-0891be3bd5156bb42 inte...	Off	10.0.1.0/24
-	subnet-02157196dk45e527e	Available	vpc-0018cbd5d1f2f153	Off	172.31.0.0/20
-	subnet-0ca10097e7b6aef64	Available	vpc-0018cbd5d1f2f153	Off	172.31.16.0/20
private-subnet-2	subnet-0e024f18717e9a592	Available	vpc-0891be3bd5156bb42 inte...	Off	10.0.4.0/24

Configured two public subnets and two private subnets across two Availability Zones within the custom VPC. Public subnets were used for the Application Load Balancer and NAT Gateway, while private subnets were used for EC2 instances.

Route Tables and NAT Gateway:



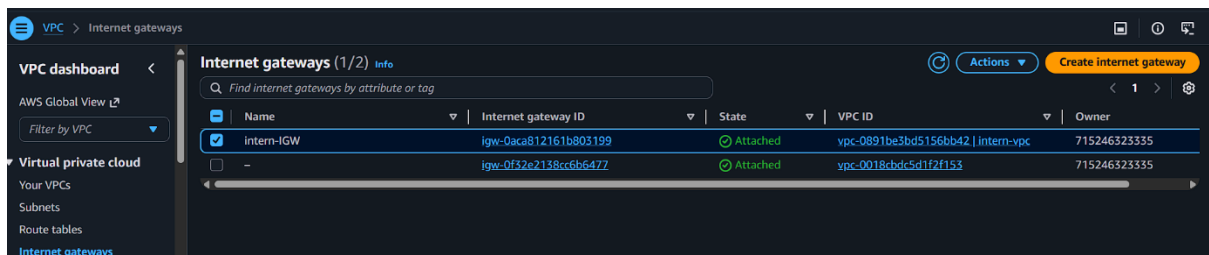
The image contains two screenshots from the AWS VPC console. The top screenshot shows the 'NAT gateways' page with one gateway named 'Intern-NAT' in an 'Available' state. The bottom screenshot shows the 'Route tables' page with three route tables: 'Public-RT' and 'Private-RT' are associated with subnets, while one is unassociated.

Name	NAT gateway ID	Connectivity...	State	State message	Availability ...	Route table ID	P
Intern-NAT	nat-0a245da3777c905fd	Public	Available	-	Zonal	-	1

Name	Route table ID	Explicit subnet associ...	Edge associations	Main	VPC
-	rtb-09b71f8ba2987f078	-	-	Yes	vpc-0891be3bd5156bb42 inte...
Public-RT	rtb-0a0d93d0bca3412e4	2 subnets	-	No	vpc-0891be3bd5156bb42 inte...
-	rtb-00de4b9fa71fd27a1	-	-	Yes	vpc-0018cbd5d1f2f153
Private-RT	rtb-01b5486cbd0661d70	2 subnets	-	No	vpc-0891be3bd5156bb42 inte...

Configured public and private route tables with Internet Gateway and NAT Gateway for internet connectivity.

Internet Gateway Configuration:

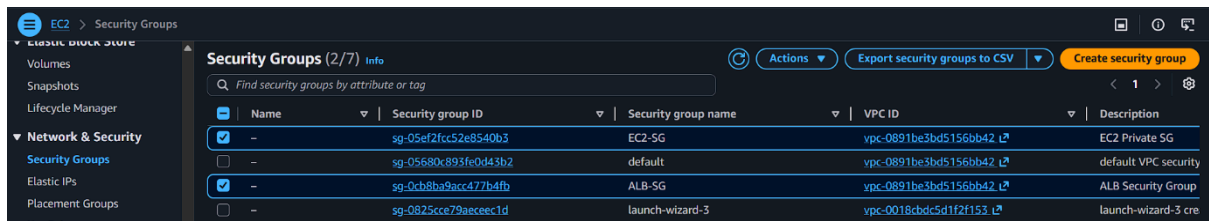


The screenshot shows the AWS VPC console's 'Internet gateways' page. The left sidebar contains the 'VPC dashboard' and 'Virtual private cloud' sections. The main area displays a table of internet gateways with columns for Name, Internet gateway ID, State, VPC ID, and Owner. Two gateways are listed: 'Intern-IGW' is attached to a VPC, and another is unattached.

Name	Internet gateway ID	State	VPC ID	Owner
Intern-IGW	igw-0aca812161b803199	Attached	vpc-0891be3bd5156bb42 intern-vpc	715246323335
-	igw-0f32e2138cc6b6477	Attached	vpc-0018cbd5d1f2f153	715246323335

An Internet Gateway was attached to the custom VPC to allow internet access for resources deployed in the public subnets.

Security Groups:

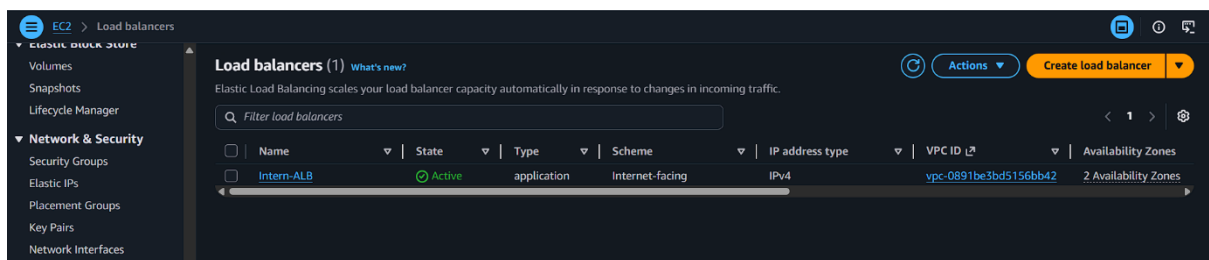


The screenshot shows the AWS Management Console 'Security Groups' page. The left sidebar has a 'Network & Security' section with 'Security Groups' selected. The main area shows a table of security groups. The first group, 'EC2-SG', is selected and has a blue checkmark. The second group, 'default', is not selected. The third group, 'ALB-SG', is selected and has a blue checkmark. The fourth group, 'launch-wizard-3', is not selected.

	Name	Security group ID	Security group name	VPC ID	Description
☒	-	sg-05ef2fcc52e8540b3	EC2-SG	vpc-0891be3bd5156bb42	EC2 Private SG
☐	-	sg-05680c893fe0d43b2	default	vpc-0891be3bd5156bb42	default VPC security
☒	-	sg-0cb8ba9acc477b4fb	ALB-SG	vpc-0891be3bd5156bb42	ALB Security Group
☐	-	sg-0825cce79aeceec1d	launch-wizard-3	vpc-0018c8dc5d1f2f153	launch-wizard-3 cre

Configured least-privilege security groups allowing HTTP traffic only from ALB to EC2 instances.

Application Load Balancer:

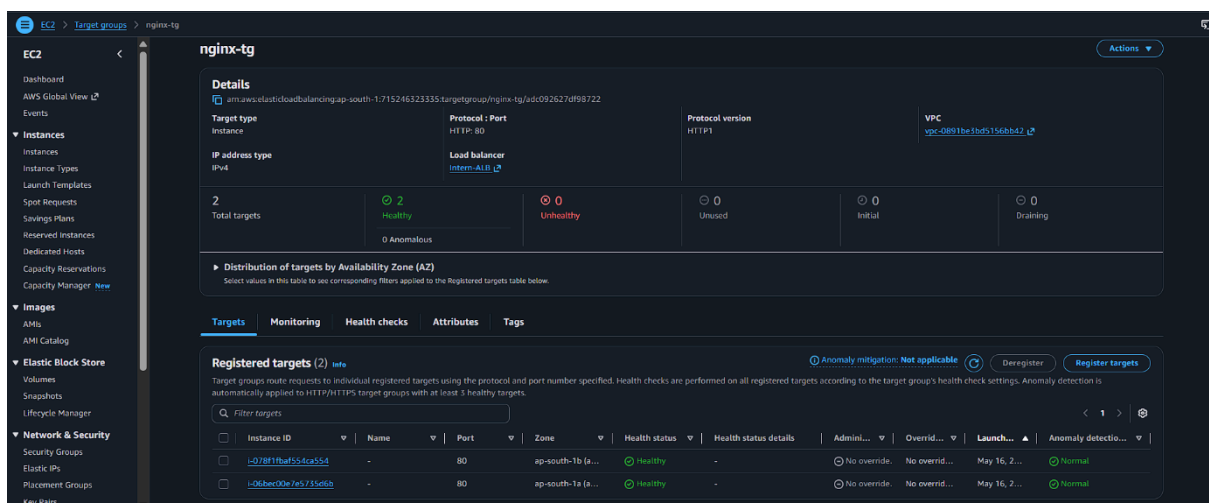


The screenshot shows the AWS Management Console 'Load balancers' page. The left sidebar has a 'Network & Security' section with 'Load balancers' selected. The main area shows a table of load balancers. The first load balancer, 'Intern-ALB', is selected and has a green checkmark. The second load balancer, 'default', is not selected.

	Name	State	Type	Scheme	IP address type	VPC ID	Availability Zones
☒	Intern-ALB	Active	application	Internet-facing	IPv4	vpc-0891be3bd5156bb42	2 Availability Zones

An Application Load Balancer was deployed across two public subnets to distribute incoming HTTP traffic across EC2 instances running in private subnets.

Target Group Healthy Instances:

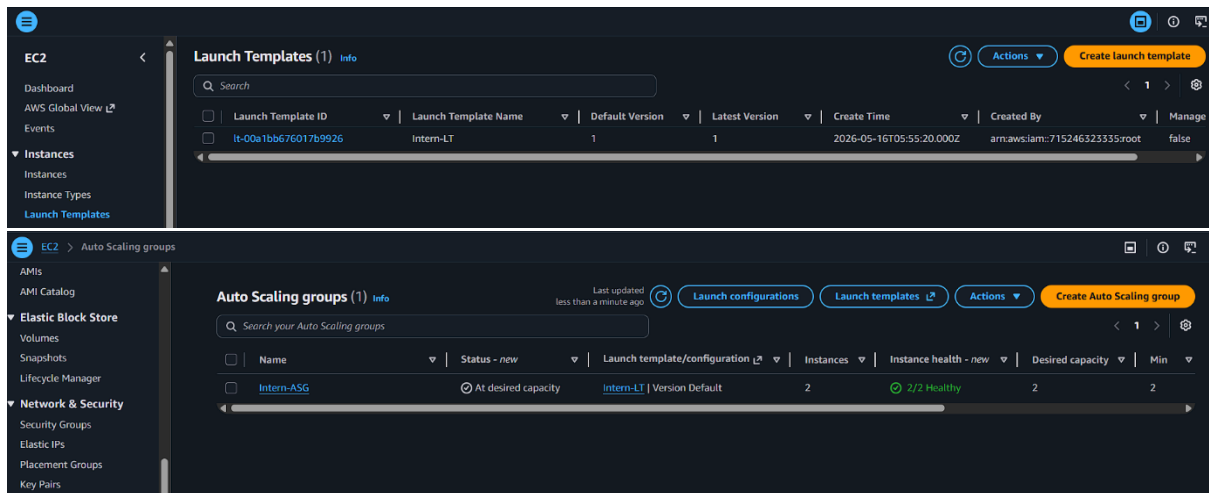


The screenshot shows the AWS Management Console 'Target Groups' page. The left sidebar has a 'Network & Security' section with 'Target Groups' selected. The main area shows the details for the 'nginx-tg' target group. The 'Details' section shows the target type as 'Instance', the protocol as 'HTTP', and the VPC as 'vpc-0891be3bd5156bb42'. The 'Targets' section shows a table of registered targets. The first target, 'i-078f1bf4554ca554', is healthy. The second target, 'i-06bec00e7e573566b', is healthy.

	Instance ID	Name	Port	Zone	Health status	Health status details	Admin...	Overrid...	Launch...	Anomaly detection...
☐	i-078f1bf4554ca554	-	80	ap-south-1b (a...	Healthy	-	No override...	No overrid...	May 16, 2...	Normal
☐	i-06bec00e7e573566b	-	80	ap-south-1a (a...	Healthy	-	No override...	No overrid...	May 16, 2...	Normal

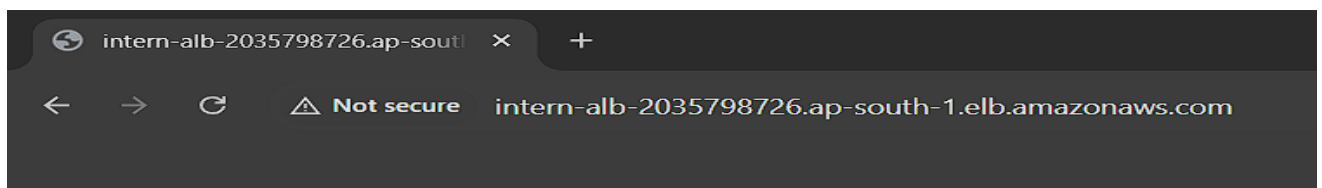
The Target Group was configured with HTTP health checks to monitor EC2 instance availability and ensure traffic was routed only to healthy instances.

Launch Template and Auto Scaling Group:



Created a Launch Template and Auto Scaling Group to automatically launch EC2 instances in private subnets.

Load Balancing Validation:



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Instance ID: i-06bec00e7e5735d6b

Availability Zone: ap-south-1a



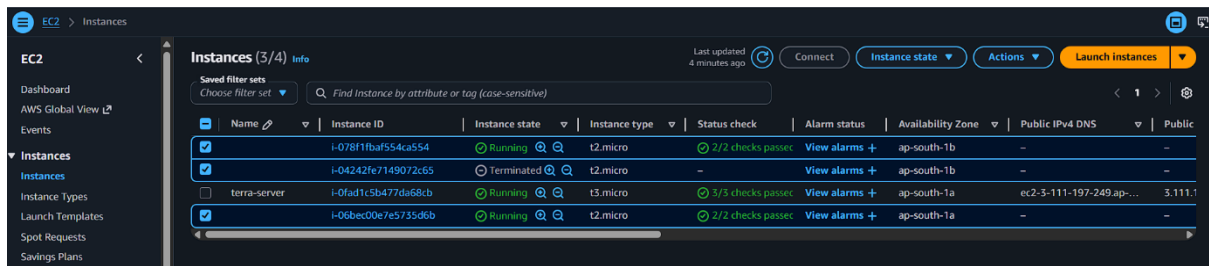
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Instance ID: i-078f1fbaf554ca554

Availability Zone: ap-south-1b

Verified load balancing by refreshing the ALB DNS and observing different instance IDs and Availability Zones.

Auto Scaling and Self-Healing Validation:



	Name	Instance ID	Instance state	Instance type	Status check	Alarm status	Availability Zone	Public IPv4 DNS	Public
☒		i-078f1fbaf554ca554	Running	t2.micro	2/2 checks passed	View alarms +	ap-south-1b	-	-
☒		i-04242fe7149072c65	Terminated	t2.micro	-	View alarms +	ap-south-1b	-	-
☐	terra-server	i-0fad1c5b477da68cb	Running	t3.micro	3/3 checks passed	View alarms +	ap-south-1a	ec2-3-111-197-249 ap-...	3.111.1
☒		i-06bec00e7e5735d6b	Running	t2.micro	2/2 checks passed	View alarms +	ap-south-1a	-	-

One EC2 instance was manually terminated. The Auto Scaling Group automatically launched a replacement instance within a few minutes, confirming self-healing capability.

Conclusion:

This project successfully demonstrated the deployment of a highly available and secure web application architecture on AWS using core cloud infrastructure services such as VPC, EC2, NAT Gateway, Application Load Balancer, and Auto Scaling Group. The implementation ensured high availability, load balancing, secure private subnet deployment, and self-healing capability across multiple Availability Zones.